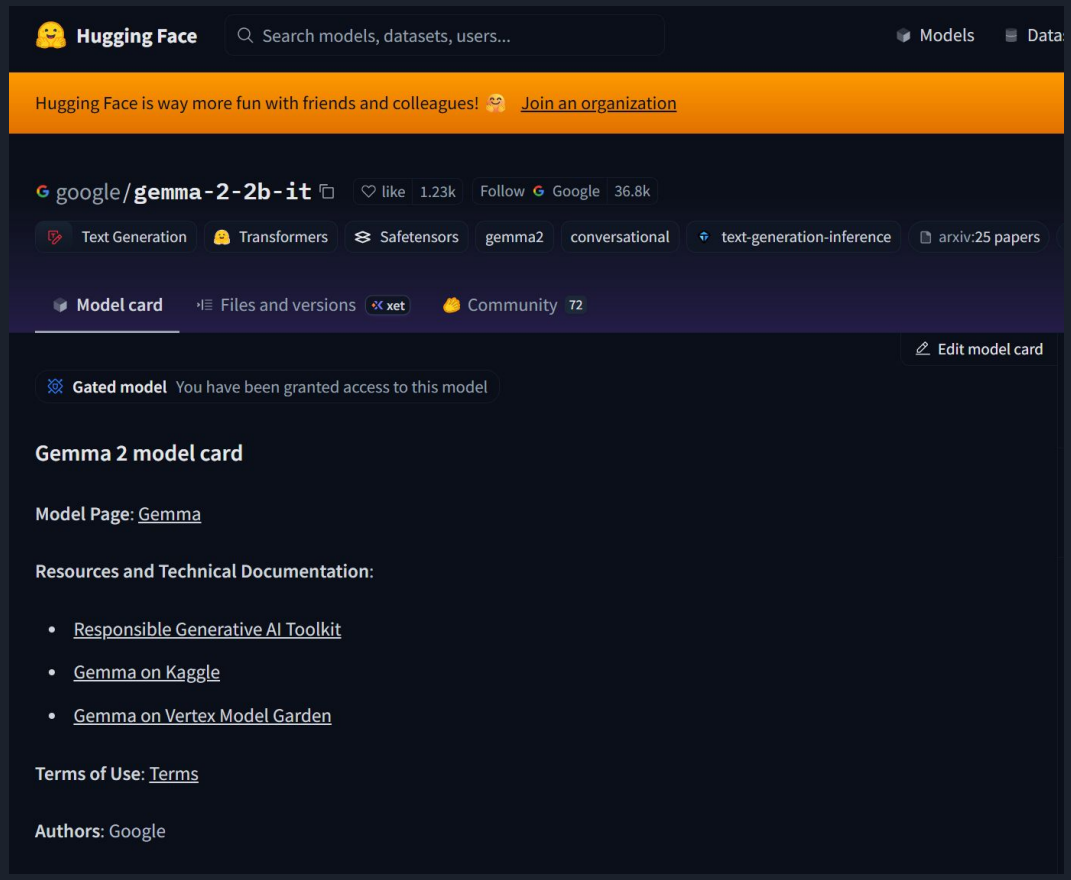
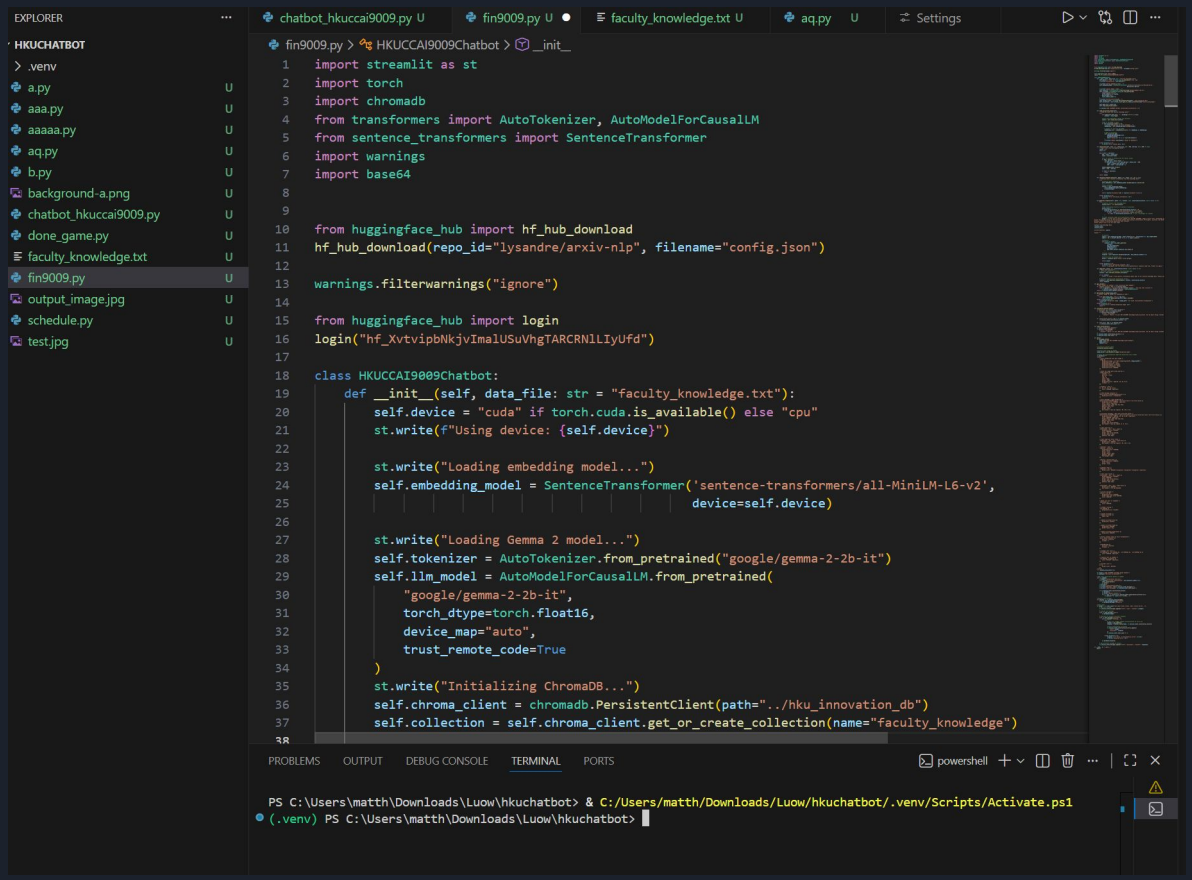
CCAI9009-Chatbot

Matthew Wong
Nabiha

Codes


Hugging Face
Search models, datasets, users...
Models
Data
Hugging Face is way more fun with friends and colleagues!
Join an organization
google/gemma-2-2b-it
like 1.23k
Follow Google 36.8k
Text Generation
Transformers
Safetensors
gemma2
conversational
text-generation-inference
arxiv:25 papers
Model card
Files and versions
xet
Community 72
Edit model card
Gated model You have been granted access to this model
Gemma 2 model card
Model Page: Gemma
Resources and Technical Documentation:
Responsible Generative AI Toolkit
Gemma on Kaggle
Gemma on Vertex Model Garden
Terms of Use: Terms
Authors: Google

Codes


EXPLORER
HKUCHATBOT
.venv
a.py U
aaa.py U
aaaaa.py U
aq.py U
b.py U
background-a.png U
chatbot_hkuccai9009.py U
done_game.py U
faculty_knowledge.txt U
fin9009.py U
output_image.jpg U
schedule.py U
test.jpg U
chatbot_hkuccai9009.py U
fin9009.py U
faculty_knowledge.txt U
aq.py U
Settings
fin9009.py > HKUCCAI9009Chatbot > __init__
import streamlit as st
import torch
import chromadb
from transformers import AutoTokenizer, AutoModelForCausalLM
from sentence_transformers import SentenceTransformer
import warnings
import base64
from huggingface_hub import hf_hub_download
hf_hub_download(repo_id="lysandre/arxiv-nlp", filename="config.json")
warnings.filterwarnings("ignore")
from huggingface_hub import login
login("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")
class HKUCCAI9009Chatbot:
def __init__(self, data_file: str = "faculty_knowledge.txt"):
self.device = "cuda" if torch.cuda.is_available() else "cpu"
st.write(f"Using device: {self.device}")
st.write("Loading embedding model...")
self.embedding_model = SentenceTransformer('sentence-transformers/all-MiniLM-L6-v2',
device=self.device)
st.write("Loading Gemma 2 model...")
self.tokenizer = AutoTokenizer.from_pretrained("google/gemma-2-2b-it")
self.llm_model = AutoModelForCausalLM.from_pretrained(
"google/gemma-2-2b-it",
torch_dtype=torch.float16,
device_map="auto",
trust_remote_code=True
)
st.write("Initializing ChromaDB...")
self.chroma_client = chromadb.PersistentClient(path="../hku_innovation_db")
self.collection = self.chroma_client.get_or_create_collection(name="faculty_knowledge")
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
PS C:\Users\matth\Downloads\Luow\hkuchatbot> & C:/Users/matth/Downloads/Luow/hkuchatbot/.venv/Scripts/Activate.ps1
(.venv) PS C:\Users\matth\Downloads\Luow\hkuchatbot>

Database

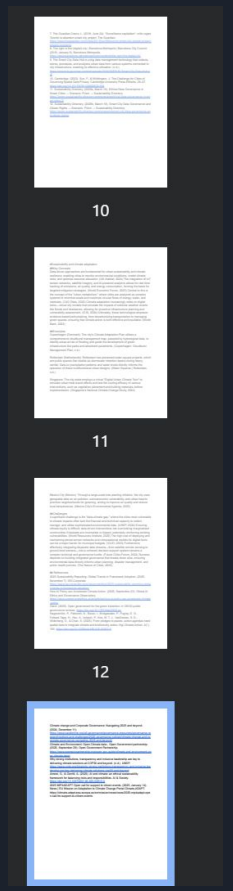
Chatbot Content Outline

These programs cover a range of topics, including smart city technologies (IoT, AI, Digital Twins), urban sensing, data analytics, GIS, and sustainable development, with a strong emphasis on both theoretical knowledge and practical, data-driven skills for future urban challenges.

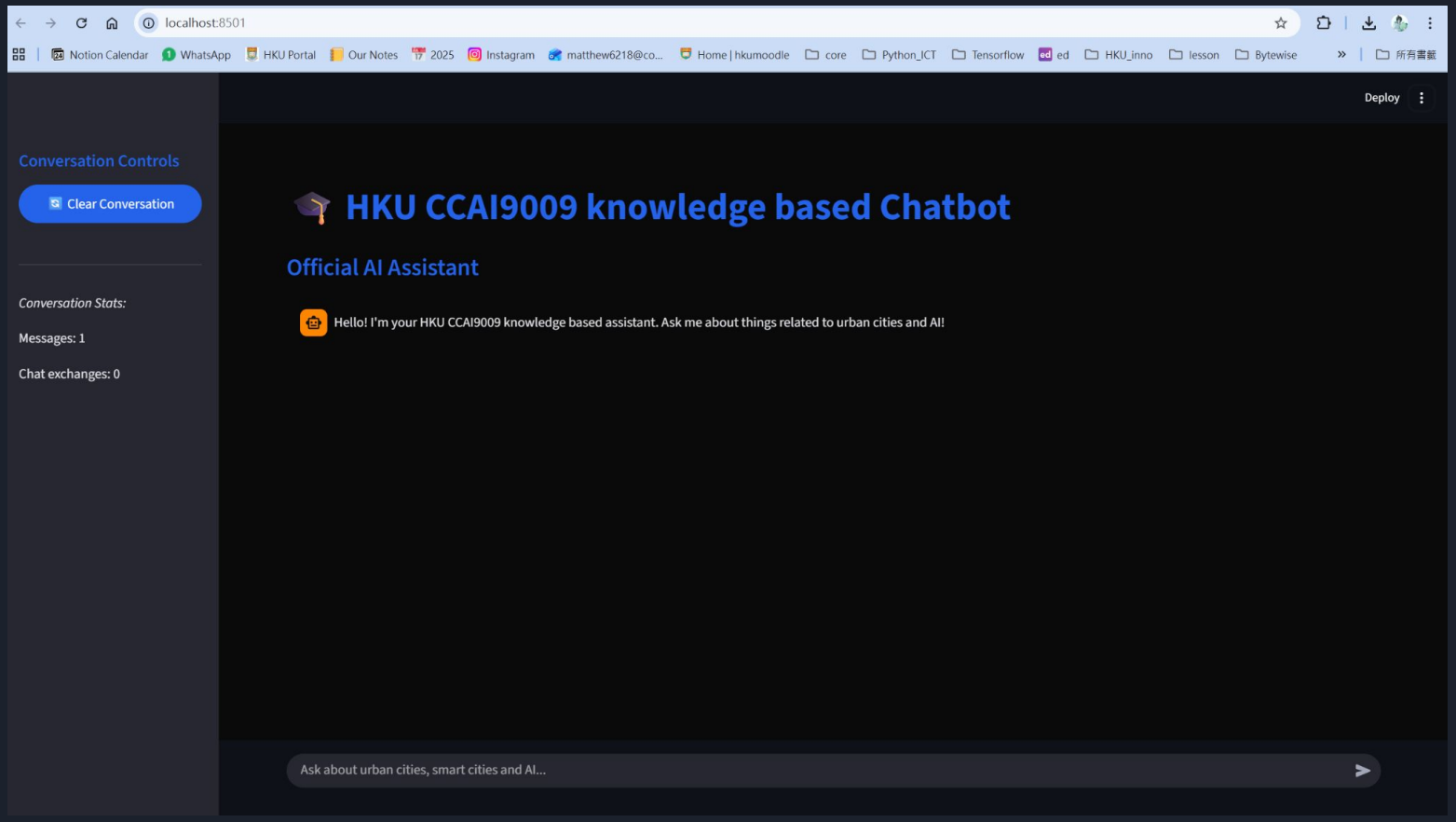
Four essential pillars of Urban Technology:

- Smart cities & IoT
- Mobility & transport systems
- Data governance & ethics
- Sustainability and climate adaptation

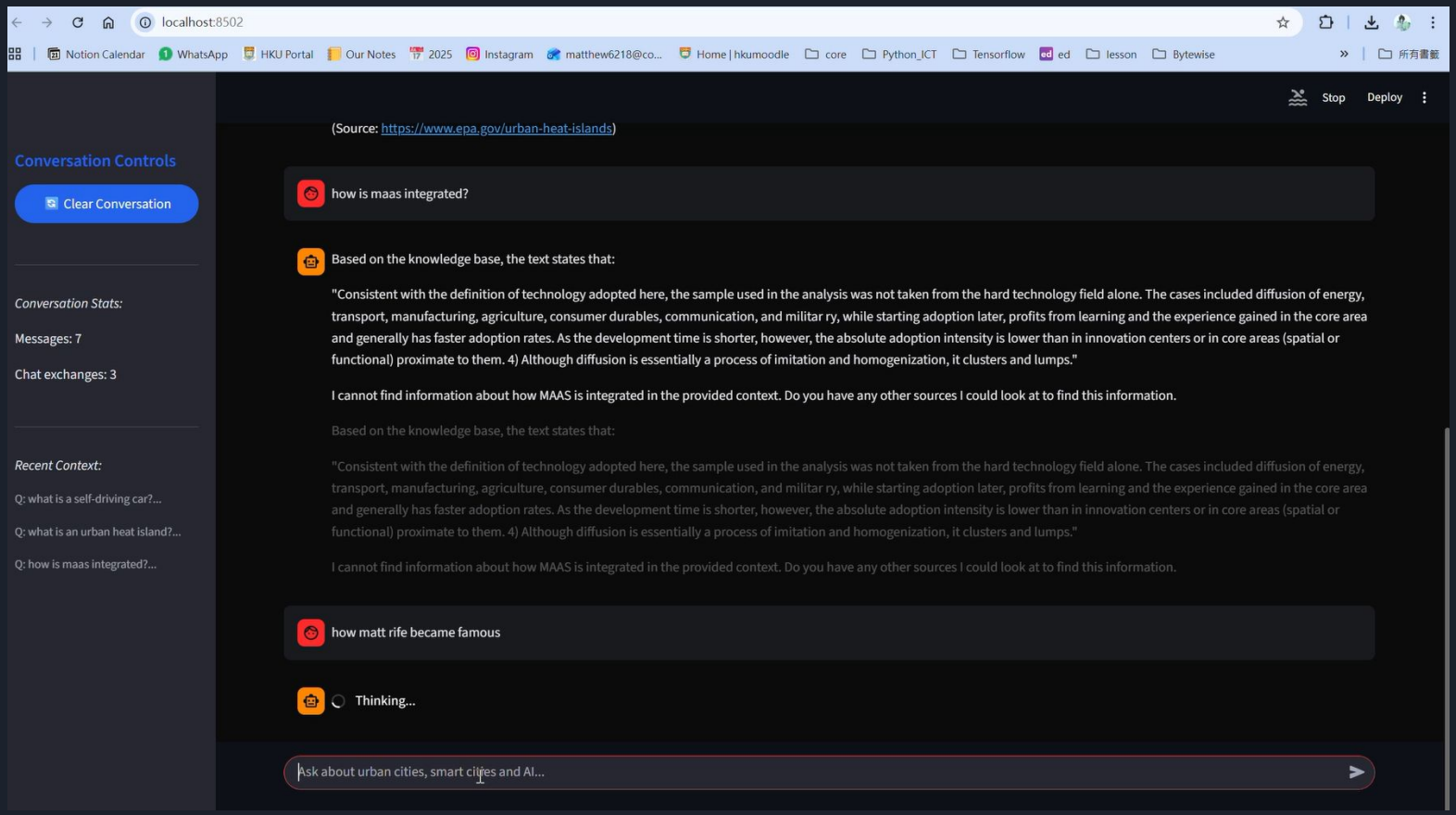
100+essay/article included for database



UI


localhost:8501
Notion Calendar
WhatsApp
HKU Portal
Our Notes
2025
Instagram
matthew6218@co...
Home | hkumoodle
core
Python_ICT
Tensorflow
ed
HKU_inno
lesson
Bytewise
所有書籤
Deploy
Conversation Controls
Clear Conversation
Conversation Stats:
Messages: 1
Chat exchanges: 0
HKU CCAI9009 knowledge based Chatbot
Official AI Assistant
Hello! I'm your HKU CCAI9009 knowledge based assistant. Ask me about things related to urban cities and AI!
Ask about urban cities, smart cities and AI...

UI


localhost:8502
Notion Calendar
WhatsApp
HKU Portal
Our Notes
2025
Instagram
matthew6218@co...
Home | hkumoodle
core
Python_ICT
Tensorflow
ed
lesson
Bytewise
所有書籤
Stop
Deploy
Conversation Controls
Clear Conversation
Conversation Stats:
Messages: 7
Chat exchanges: 3
Recent Context:
Q: what is a self-driving car?...
Q: what is an urban heat island?...
Q: how is maas integrated?...
(Source: https://www.epa.gov/urban-heat-islands)
how is maas integrated?
Based on the knowledge base, the text states that:
"Consistent with the definition of technology adopted here, the sample used in the analysis was not taken from the hard technology field alone. The cases included diffusion of energy, transport, manufacturing, agriculture, consumer durables, communication, and militar ry, while starting adoption later, profits from learning and the experience gained in the core area and generally has faster adoption rates. As the development time is shorter, however, the absolute adoption intensity is lower than in innovation centers or in core areas (spatial or functional) proximate to them. 4) Although diffusion is essentially a process of imitation and homogenization, it clusters and lumps."
I cannot find information about how MAAS is integrated in the provided context. Do you have any other sources I could look at to find this information.
how matt rife became famous
Thinking...
Ask about urban cities, smart cities and AI...

Poster


Discover the Future of Urban Technology
URBANTECH EXPLORER CHATBOT
Want to Know How Cities Are Becoming Smarter? Ask me!
Smart Cities & IoT
How sensors, apps, and data connect everything from traffic lights to energy systems.
Mobility & Transport Systems
From AI-powered trains to bike cities and the future of electric transport.
Data Governance & Ethics
Who owns city data? How do we protect privacy in a connected world?
Sustainability & Climate Adaptation
How cities fight heatwaves, flooding, and pollution with green innovation.
TRENDING NOW
Blockchain for cities
Digital twins
Circular economy
Smart Lighting
Green Architecture
UNLOCK THE FUTURE
Bite-sized explanations for beginners
Updated with the latest ideas shaping cities
Perfect for quick facts or fun learning
TEAM: NABIHA (3036668298)
MATTHEW (3036582808)
CCAI9009 AI Chatbot Project
POWERED BY OLLAMA
Fast answers. Big ideas. Smart cities made simple.